

- EMD 2021 -

Exo 1: 1) 2 max ?

$$D_{\max} = n \cdot R_{\max} \quad (0,25)$$

Trouvons la valence du signal à partir du graphe:

$$\begin{cases} 2A : A_1, A_2 \\ 2 \text{ frequences} : f, 2f \\ \varphi = \pi/2, 0, 3\pi/2, \pi \end{cases}$$

$$V = 2A * 2f * 4\varphi = 16 = 2^4 \\ \Rightarrow n = 4 \quad (0,5)$$

$$L_{\max} = \frac{D_{\max}}{n} = \frac{9600}{4} = 2400 \text{ bands} \quad (0,5)$$

$$R_{\max} = \frac{1}{D_{\max}} \Rightarrow D_{\max} = \frac{1}{R_{\max}} = \frac{1}{2400} = 0,0004166 \\ \approx 0,42 \cdot 10^{-3} \text{ s}$$

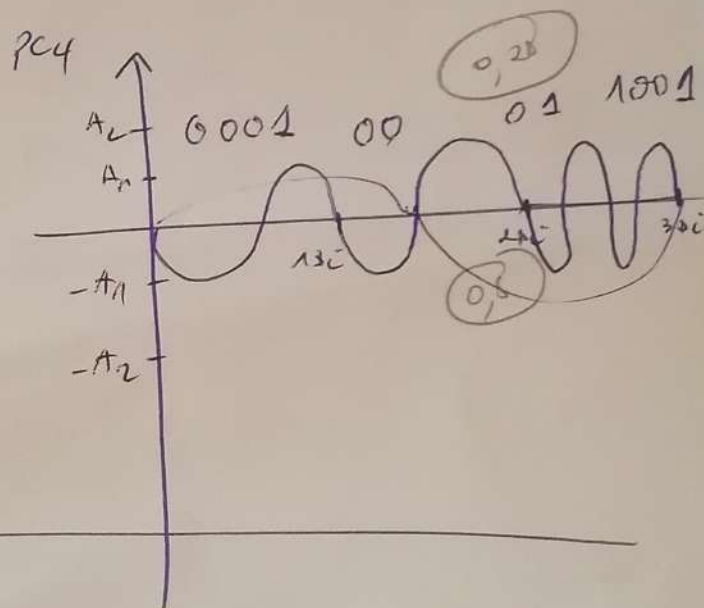
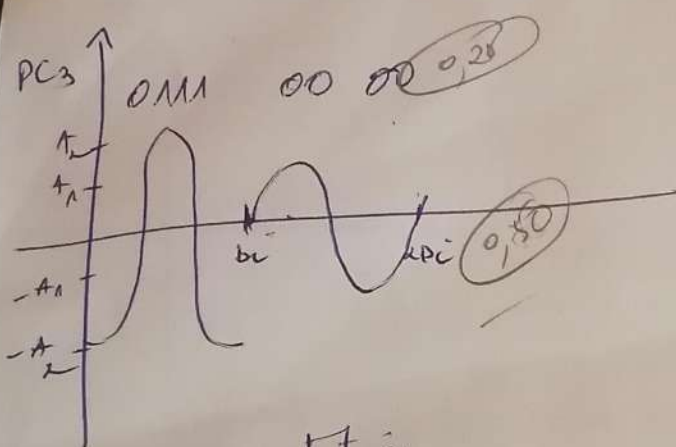
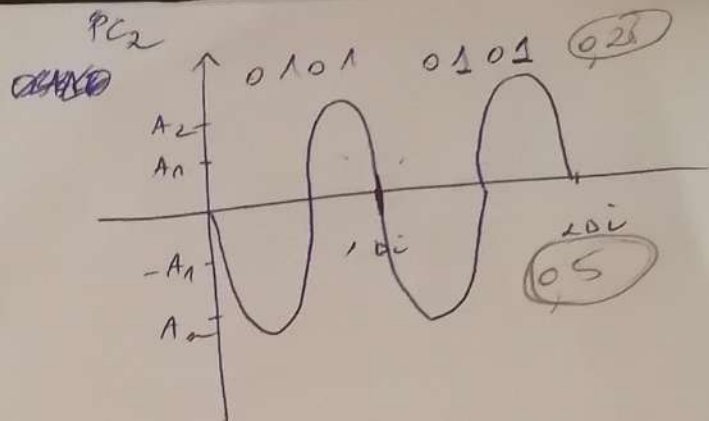
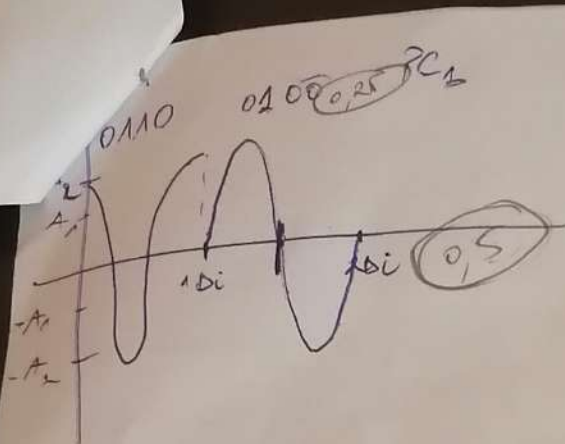
$$D_{\max} \approx 0,42 \text{ ms} \quad (0,25)$$

$$2) \quad Q = D_{\max} + \frac{D_{\max}}{2} = 0,42 + \frac{0,42}{2} = 0,42 + 0,21 = 0,63 \text{ ms} \\ Q = 0,63 \text{ ms} \quad (0,5)$$

$$1 D_{\max} \Rightarrow 4 \text{ bits} ; \quad 1 D_{\max} + \frac{D_{\max}}{2} \Rightarrow 6 \text{ bits} \quad (0,5)$$

$$3) \quad D_i = \frac{1}{R_i} = R_i = \frac{R_{\max}}{4} = 600 \text{ bands} \quad (0,5)$$

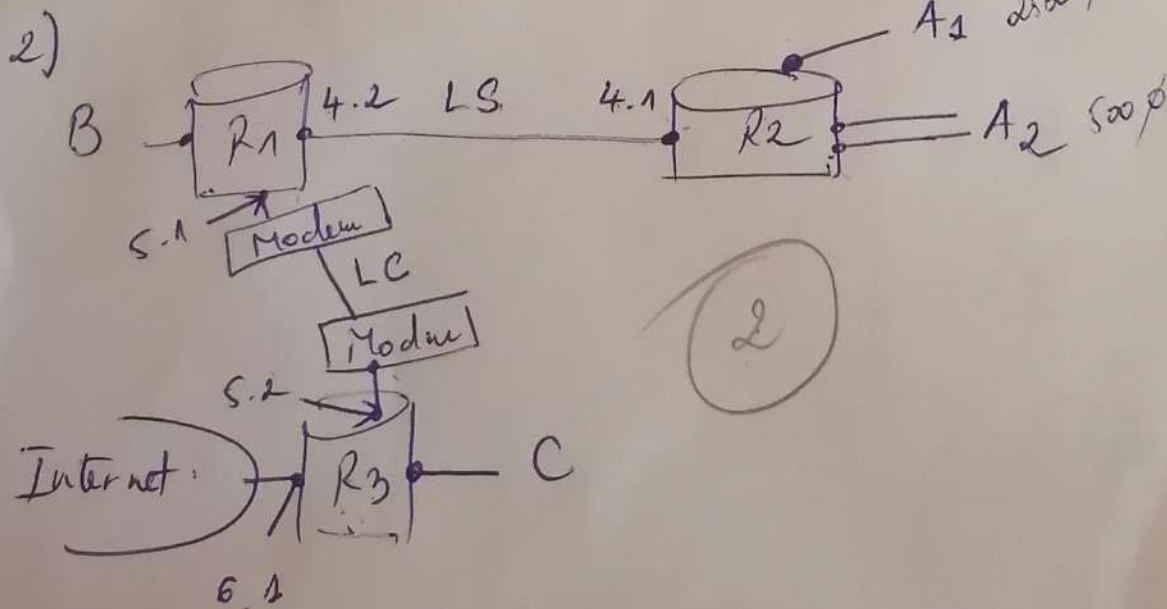
$$D_i = \frac{1}{600} = 1,66 \text{ ms} \quad D_i = 1,66 \text{ ms} \quad (0,25)$$



4) 2 simulations

Exo 2:

1) Pont car A_1 et A_2 ont 2 méthodes d'accès $\neq$
 Bus IRR702.3 et Token Ring IRR702.5
 sur Ethernet



155.50.128.0 /16

Etude des besoin = Nb SR = 7SR $\Rightarrow$ 3 bits SR

au Max 50 machines

Les masque à partir /19
jusqua /25

A1: 155.50.0.0 /24

A2: 155.50.1.0 /24

B: 155.50.2.0 /24

C: 155.50.3.0 /24

LS: 155.50.4.0 /24

LC: 155.50.5.0 /24

ADSL: 155.50.6.0 /24

4)

R1

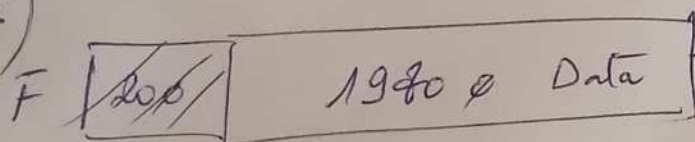
R2

R3

R1	sortie	R2	sortie	R3	sortie
A1	155.50.4.1	directe	155.50.5.1		
A2	155.50.4.1	directe	"		
B	directe	155.50.4.2	"		
C	155.50.5.2	155.50.4.2	directe		
Internet	155.50.5.2	155.50.4.2	directe		

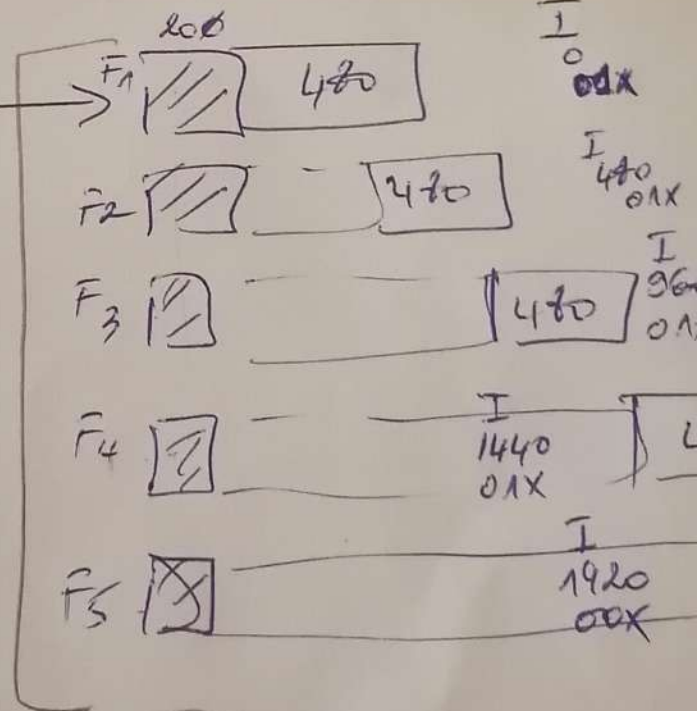
+ 1-pt interfaces

5)



R2: Fragmentation

0,25



0,25 * 5

x03:

(4)

2)

"O"

"S"

"I"

(LRC)

OSI(VRC/LRC)

= 10011111 10100110

10000111 10111110

LRC

(01)

(15)

1	1	1	1
0	0	0	0
0	1	0	1
1	0	0	1
1	0	0	1
1	1	1	1
1	1	1	1

VRC

1	0	1	0
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2)

(1)

1	1	1	1
0	0	0	0
0	1	0	1
1	0	1	1
1	0	0	1
1	1	1	1
1	1	1	1
1	0	1	0

Il est mal regu

Où, en inversant le bit de 3^{me} colonne et 4^{me} ligne.

(1)